

GazeSpeak: A Low-Cost, LLM-Accelerated Eye Gaze Typing System for ALS Patients

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Abstract—Amyotrophic Lateral Sclerosis (ALS) progressively eliminates voluntary muscle control while typically sparing eye movement, making gaze-based text entry one of the few remaining communication channels for patients in advanced stages of the disease. Existing eye-gaze augmentative and alternative communication (AAC) systems are often slow, expensive, and dependent on dedicated infrared hardware. This paper presents GazeSpeak, an open-source, webcam-only assistive eye gaze typing tool that combines MediaPipe-based iris tracking, three-point horizontal calibration, a step-based gaze-navigation scheme, large language model (LLM)-powered word prediction and abbreviation expansion, an LLM-driven binary-choice conversation engine, rapid-blink emergency alerting with SMS notification, and text-to-speech output. The system runs on consumer hardware using a single webcam and requires no infrared tracker. We describe the architecture, implementation, and design rationale of each module, and situate the work within recent literature on LLM-accelerated gaze typing, calibration-free interfaces, and brain-computer interface typing paradigms for ALS. GazeSpeak is implemented in Python using PyQt6, OpenCV, and MediaPipe, with Groq-hosted Llama 3.3 70B providing LLM capabilities. The system is designed to operate across multiple stages of ALS progression through its modular, multi-modal input architecture.

Index Terms—Amyotrophic Lateral Sclerosis, Eye Gaze Typing, Augmentative and Alternative Communication, Large Language Models, Assistive Technology, MediaPipe, Webcam-Based Tracking.

I. INTRODUCTION

Amyotrophic Lateral Sclerosis (ALS) is a progressive neurodegenerative disease that destroys motor neurons, gradually stripping patients of the ability to speak, move their limbs, and eventually breathe independently [3], [9]. Because the

oculomotor pathways are typically preserved even in advanced disease stages, eye gaze has become the primary input modality for augmentative and alternative communication (AAC) systems used by this population [1], [6].

Despite decades of research, gaze typing remains slow. On-screen keyboards driven by eye trackers typically support fewer than ten words per minute, roughly an order of magnitude below conversational speech [6]. This gap discourages extended interaction and contributes to the social isolation commonly reported among ALS patients [8]. Two additional obstacles compound the problem. First, many gaze-tracking systems depend on dedicated infrared hardware and lengthy calibration procedures that are expensive, difficult for patients with restricted head control, and prone to drift [2], [5], [14]. Second, research on typing acceleration, low-cost tracking, brain-computer interfaces (BCI), and voice preservation has advanced along mostly separate paths, with comparatively little work integrating these advances into a single deployable tool tailored to ALS [1].

Three specific advances motivate this work. LLM-powered interfaces have demonstrated 29–60% improvements in text-entry rate over baseline keyboards and 57% savings in motor actions [1]. Blink-coded alphabet trees have enabled calibration-free typing on ordinary webcams [2]. Webcam-based deep-learning facial landmark trackers have removed the dependence on infrared hardware entirely [5]. BCI paradigms offer a fallback modality for patients who progress to a fully locked-in state [3], [11].

This paper presents GazeSpeak, an open-source assistive eye gaze typing system that integrates several of these advances

into a working, deployable application. The system runs on consumer hardware with a single webcam, uses MediaPipe Face Mesh for iris tracking, provides LLM-accelerated word prediction and abbreviation expansion through Groq-hosted Llama 3.3 70B, supports a binary-choice conversation mode for patients who find letter-by-letter typing too slow, and includes a rapid-blink emergency alert system with SMS notification via Twilio. The specific contributions are:

- 1) A complete, open-source implementation of a webcam-only gaze typing system that requires no infrared hardware or commercial eye tracker.
- 2) A three-point horizontal calibration scheme with IQR-based outlier rejection that adapts gaze zone thresholds to each patient's eye movement range.
- 3) A step-based gaze navigation model with hysteresis-filtered zone transitions, where horizontal eye direction acts as discrete left/right arrow keys rather than continuous cursor control.
- 4) Integration of LLM-based word prediction, context-aware abbreviation expansion, and a binary-choice conversation engine that reduces the motor effort required for communication.
- 5) A rapid-blink emergency alert mechanism using Eye Aspect Ratio (EAR) blink detection, audible alarm, and Twilio SMS notification.

II. RELATED WORK

A. LLM-accelerated gaze typing

Cai et al. developed SpeakFaster, an LLM-powered interface for abbreviated text entry that saved 57% more motor actions than traditional predictive keyboards in offline simulation and achieved text-entry rates 29–60% above baseline in lab and field testing with two eye-gaze AAC users with ALS [1]. A companion field study instrumented the everyday gaze-typed communication of an ALS patient over an extended period to characterize real-world conversational patterns [6]. Caria surveyed the convergence of BCI spellers with LLMs, tracing the evolution from early statistical language models to contemporary transformer-based predictive writing [7]. Mirbagheri and Chau presented EyeChat, a gaze-controlled web-browsing and chatbot interface for an augmented-reality display, reporting 9.88 words per minute with an LLM-driven response generation system [16].

B. Low-cost webcam and blink-based interfaces

Lee and Lee proposed a blink-to-alphabet tree that narrows a 26-letter alphabet through three sequential blink-coded decisions and integrates an LLM API for error correction, outperforming three baseline methods in speed and accuracy without calibration [2]. Karatay et al. developed a webcam-only, deep-learning-based facial landmark tracker that eliminates the need for blinking, enabling use by patients who can no longer blink voluntarily [5]. Molina-Cantero et al. reviewed visible-light, low-cost camera eye-tracking methods, cataloguing calibration strategies, head-pose invariance techniques, and the persistent trade-off between affordability and tracking accuracy [14].

C. Brain-computer interface typing

Li et al. conducted a PRISMA-guided review of 23 studies applying BCI paradigms to ALS communication, concluding that recent multidisciplinary efforts have improved decoding precision but that user variability, BCI illiteracy, and fatigue remain open limitations [3]. Peters et al. introduced Inquiry Preview, a multimodal extension of the RSVP-Keyboard that fuses switch input with EEG and language-model evidence; a potential end user with incomplete locked-in syndrome typed fastest with the combined modality [11]. An NIH-funded initiative on a rapid neural typing interface for veterans with ALS reflects continued investment in translating BCI-based typing toward deployment [15].

D. Datasets, hardware, and broader communication support

Ngo et al. released EEGET-ALS, a public EEG and eye-tracking dataset from six ALS patients and 170 healthy controls performing spelling tasks [4]. Liu et al. demonstrated a wireless MXene-based smart contact lens that encodes blink patterns into a resonance-frequency signal, validating biocompatibility in vivo [13]. On the communication support side, Garbey et al. developed AI tools combining pulse and speech analysis to detect emotional responsiveness during ALS clinic visits [8]. Regondi et al. combined voice banking with AI voice-cloning to preserve a patient's vocal identity for AAC devices [9]. Barton et al. surveyed AAC specialists on tele-AAC versus in-person evaluations for ALS patients [10]. A medRxiv study proposed an AI-driven approach to augmentative and assistive communication for ALS patients [12].

III. SYSTEM ARCHITECTURE

GazeSpeak is a desktop application built in Python. The system is organized into five modules: gaze tracking, user interface, text prediction, emergency alerts, and audio. Fig. 1 illustrates the data flow between these modules. The application is built on PyQt6 for the GUI, OpenCV for webcam capture, MediaPipe Face Mesh for facial landmark detection, and Groq-hosted Llama 3.3 70B for LLM capabilities.

A. Gaze tracking

The gaze tracker runs as a background QThread, capturing webcam frames at 640×480 resolution and processing each frame through MediaPipe's Face Landmarker task. The model detects 478 facial landmarks per frame, including five iris landmarks per eye (indices 468–477). Gaze direction is computed as the horizontal and vertical iris ratio: the position of each iris center relative to the corresponding eye corners, normalized to $[0, 1]$. The ratios from both eyes are averaged to produce a single gaze coordinate.

A confidence metric is computed from the agreement between the two eyes. When iris landmarks are unavailable (e.g., due to occlusion), a fallback method estimates gaze from eye corner midpoints at reduced confidence. An exponential moving average (EMA) filter smooths the gaze signal before it reaches the navigation controller. The smoothing factor is

user-configurable, defaulting to 0.25, which balances responsiveness against jitter.

B. Three-point horizontal calibration

On each application launch, a full-screen calibration sequence records the patient’s gaze ratio at three horizontal positions: fully left, straight ahead (center), and fully right. At each position, the system collects 60 gaze samples (approximately two seconds at 30 fps), rejects outliers using IQR filtering (samples outside $Q_1 - 1.5 \cdot \text{IQR}$ to $Q_3 + 1.5 \cdot \text{IQR}$ are discarded), and takes the mean of the remaining samples.

The three recorded values (x_{left} , x_{center} , x_{right}) are used to define zone boundaries. The calibrated screen position (screen_x) of a raw gaze reading x_{raw} is given by Eq. (1):

$$x_{\text{screen}} = \frac{x_{\text{raw}} - x_{\text{left}}}{x_{\text{right}} - x_{\text{left}}} \quad (1)$$

The center point in calibrated space, $c = (x_{\text{center}} - x_{\text{left}}) / (x_{\text{right}} - x_{\text{left}})$, is then used to set asymmetric dead-zone thresholds (zone_left and zone_right), as shown in Eq. (2) and Eq. (3):

$$z_{\text{left}} = \max(0, c - \max(0.55c, 0.04)) \quad (2)$$

$$z_{\text{right}} = \min(1, c + \max(0.45(1 - c), 0.12)) \quad (3)$$

The asymmetric margins and minimum floors ensure both left and right zones are always reachable regardless of where the patient’s neutral gaze falls. Calibration data is validated: the spread between left and right must exceed 0.05 in raw gaze units, and the center value must lie strictly between them. If validation fails, the patient is asked to retry.

C. Step-based gaze navigation with hysteresis

Rather than mapping gaze position to a continuous screen cursor, GazeSpeak uses a three-zone, step-based navigation model. Looking left of z_{left} triggers a single leftward step of the highlight. Looking right of z_{right} triggers a single rightward step. Returning to the center zone (between the two thresholds) stops navigation and arms dwell-time selection: if the patient’s gaze remains centered for a configurable dwell period (default 1200 ms), the currently highlighted key is selected.

To prevent false triggers from noisy gaze data, zone transitions require confirmation over multiple consecutive frames (hysteresis). A zone change is only committed after two consecutive frames (at 30 fps, approximately 66 ms) indicate the same new zone. This filters out transient spikes without adding perceptible delay. The navigation highlight wraps circularly through prediction suggestions and keyboard rows. When the highlight moves past the last key in the last keyboard row, it wraps to the first prediction slot, and vice versa.

D. On-screen keyboard and dwell selection

The keyboard is a QWERTY layout augmented with special keys: SPACE, BACKSPACE, SPEAK (triggers TTS), CLEAR, PHRASES (switches to the quick-phrases panel), and SETTINGS. Key selection occurs through dwell: when the patient returns gaze to the center zone after navigating to the desired

key, a countdown timer begins. A visual progress ring fills around the highlighted key. If gaze remains centered for the full dwell duration, the key is pressed. Any zone transition cancels the timer.

E. Word prediction

GazeSpeak combines two prediction paths running in parallel. *Local prefix matching*: a frequency-ranked dictionary (approximately 300 entries weighted toward medical, comfort, and daily-need vocabulary) provides instant completions as the patient types. Results are sorted by frequency and filtered to exclude exact matches with the current prefix. *LLM context-aware prediction*: in the background, the current sentence and partial word are sent to Groq-hosted Llama 3.3 70B via the Groq API. The model is prompted to return a JSON array of the five most likely word completions, prioritizing ALS patient communication context (medical needs, comfort, social interaction). When the LLM response arrives, it replaces the local predictions in the UI. Temperature is set to 0.3 for focused predictions.

The two-tier approach ensures the patient always sees suggestions instantly (local), while benefiting from contextually smarter predictions when the LLM response returns (typically within 200–500 ms).

F. Context-aware abbreviation expansion

For patients who prefer to type shorthand, the system detects likely abbreviations using a set of heuristics: the input is short (≤ 5 characters), contains few or no vowels relative to consonants, and does not match a list of common English short words. When an abbreviation is detected, the system sends it along with the full conversation history (caretaker questions and prior patient responses) to Llama 3.3 70B, which returns 2–4 plausible expansions ranked by confidence.

For example, if the caretaker asks “Where did the sound come from?” and the patient types “ftk”, the system might suggest “from the kitchen”, “from the kids”, or “from the terrace”. Each letter in the abbreviation is treated as potentially the first letter of a word. The LLM response includes confidence scores and reasoning for each expansion, and the suggestions are ranked by confidence before display.

G. Binary-choice conversation engine

For situations where letter-by-letter typing is too slow or fatiguing, GazeSpeak provides a conversation mode. When the caretaker types or speaks a question (via the context bar at the bottom of the screen), the system generates exactly two answer options using Llama 3.3 70B. The options are displayed as two large cards: one on the left side and one on the right side of the screen.

The patient selects an option by looking in the corresponding direction (left or right gaze). After each selection, the engine generates a follow-up pair that narrows the patient’s intended meaning. After up to five rounds of binary narrowing, the LLM composes a natural spoken sentence from the patient’s selection trail and speaks it aloud via TTS. For

example, if the caretaker asks “How are you feeling?”, the patient might narrow through “Not great” → “I have pain” → “In my head” across three rounds, and the system speaks the composed sentence “I’m not great. I have pain in my head.”

This mode requires only three gaze gestures to compose a full response, compared to dozens of individual key selections for letter-by-letter typing. The patient can also request regeneration of options, undo a previous selection, or switch to keyboard mode at any point.

H. Rapid-blink emergency alert

The gaze tracker continuously computes the Eye Aspect Ratio (EAR) for both eyes using six landmark points per eye, as shown in Eq. (4):

$$\text{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2 \|p_1 - p_4\|} \quad (4)$$

where p_1 – p_6 are the outer corner, upper lid (two points), inner corner, and lower lid (two points) respectively. A blink is registered when the average EAR across both eyes drops below 0.21 for at least two consecutive frames and then rises above the threshold (i.e., the eyes reopen).

The BlinkAlertManager maintains a sliding window of blink timestamps. If five or more blinks occur within a four-second window, the system triggers an emergency alert: (1) a full-screen red overlay covers the application; (2) a continuous siren alarm plays through the system speaker (alternating frequency sweeps between 800–1200 Hz); (3) an SMS is sent to the configured caregiver phone number via the Twilio API. A 60-second cooldown prevents repeat triggers. Blink detection is automatically paused during calibration and resumes with a two-second grace period to avoid false positives from residual calibration blinks.

I. Caregiver mode and speech-to-text input

The caregiver interacts through a context bar at the bottom of the screen. The caregiver can type a question directly via the physical keyboard or use speech-to-text (AssemblyAI real-time streaming API) to dictate their question verbally. The transcribed or typed question is then fed to both the abbreviation expander (as context) and the conversation engine (as the prompt for binary-choice generation). The application also supports full keyboard and mouse input for caregiver-side configuration, settings adjustment, and recalibration.

J. Text-to-speech output

Composed sentences, quick phrases, and conversation-mode responses are spoken aloud using the pyttsx3 offline TTS engine. Speech rate is user-configurable through the settings panel.

IV. IMPLEMENTATION DETAILS

Table I summarizes the technology stack. The application is open-source and runs on Windows with Python 3.10+.

The codebase is organized into five Python packages: *gaze/* — gaze tracker (background QThread) and calibration screen; *ui/* — nine PyQt6 widget modules (keyboard, gaze

TABLE I
TECHNOLOGY STACK

Layer	Library	Version
GUI framework	PyQt6	≥6.6.0
Webcam capture	OpenCV	≥4.8.0
Face/iris tracking	MediaPipe	≥0.10.9
Numerical computation	NumPy	≥1.24.0
LLM inference	Groq (Llama 3.3 70B)	≥1.0.0
Text-to-speech	pyttsx3	≥2.90
Emergency SMS	Twilio	≥9.0.0
Speech-to-text	AssemblyAI	≥0.30.0

TABLE II
FEATURE COMPARISON WITH SURVEYED SYSTEMS

Feature	SF [1]	B2A [2]	WCI [5]	GS
Webcam-only	×	✓	✓	✓
LLM prediction	✓	✓	×	✓
Abbreviation expansion	✓	×	×	✓
Conversation mode	×	×	×	✓
Emergency alert	×	×	×	✓
Caregiver input	×	×	×	✓
Calibration-free	×	✓	×	×
Open-source	✓	N/A	N/A	✓

SF = SpeakFaster, B2A = Blink-to-Alphabet Tree, WCI = Webcam Interface, GS = GazeSpeak (this work).

cursor, sentence bar, prediction bar, quick phrases, settings panel, context bar, conversation panel, and emergency screen); *prediction/* — word predictor (local + LLM), abbreviation expander, conversation engine, and the frequency-ranked dictionary; *alerts/* — rapid-blink detection and Twilio SMS emergency alert system; *audio/* — AssemblyAI-based real-time speech-to-text for caregiver input.

Threading model: the gaze tracker runs as a dedicated QThread, emitting Qt signals for gaze updates, frame readiness, tracking loss, and blink events. LLM calls (word prediction, abbreviation expansion, conversation options, and response composition) run in daemon threads to avoid blocking the GUI. Results are marshalled back to the main thread via `QTimer.singleShot(0, ...)` or a dedicated `pyqtSignal`.

The system operates without an internet connection for basic gaze typing, dwell selection, local word prediction, TTS, and emergency alarm. Internet connectivity is required only for LLM-based prediction, abbreviation expansion, conversation mode, SMS alerts, and speech-to-text.

V. DISCUSSION

A. Relation to prior work

Table II compares GazeSpeak’s feature set with the systems described in the surveyed literature.

GazeSpeak is, to our knowledge, the first system to combine webcam-only iris tracking, LLM-accelerated prediction, context-aware abbreviation expansion, a binary-choice conversation engine, and emergency blink alerting in a single open-

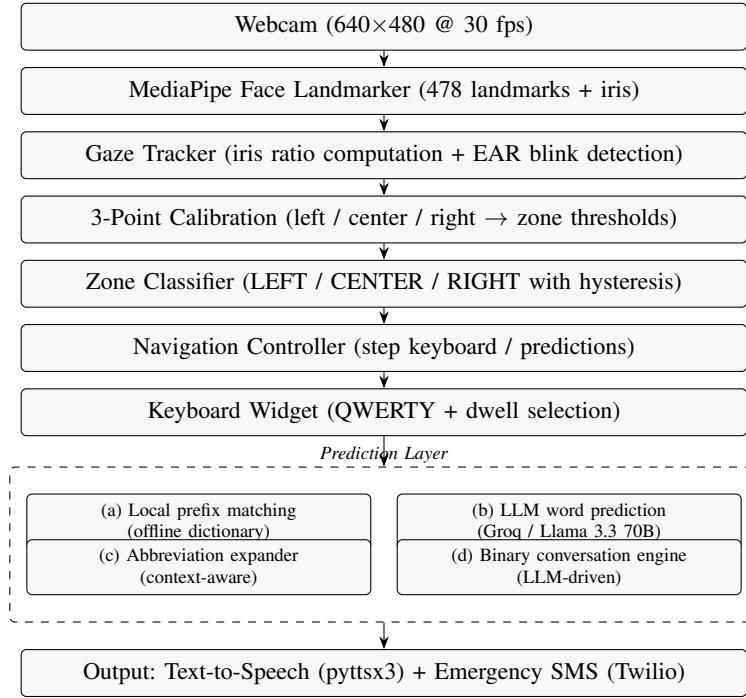


Fig. 1. Data flow through the GazeSpeak pipeline.

source tool for ALS patients. SpeakFaster [1] demonstrated that LLM acceleration reduces motor actions by 57%; GazeSpeak applies a similar principle using Groq-hosted Llama 3.3 70B for both word prediction and abbreviation expansion. The blink-to-alphabet tree [2] showed that calibration-free typing is feasible with LLM error correction; while GazeSpeak still requires a brief calibration, the three-point procedure takes approximately ten seconds and adapts to each patient’s range.

The binary-choice conversation engine is a contribution not present in any of the surveyed systems. It reduces a full sentence response to three to five gaze gestures, which is particularly useful for patients whose fatigue limits sustained letter-by-letter typing. The emergency blink alert addresses a safety concern raised in multiple care-focused studies [8], [10]: patients who lose the ability to call for help need a hands-free, gaze-compatible alarm mechanism.

B. Design decisions

Several design choices warrant discussion.

Step-based versus continuous cursor. Continuous gaze-to-cursor mapping is the approach used by most commercial eye trackers [14]. We chose step-based navigation because webcam-only iris tracking lacks the spatial precision of infrared systems. By discretizing gaze into three zones, the system tolerates tracking noise that would make continuous cursor control impractical. The trade-off is slower maximum throughput, since each key requires at least one zone transition.

Hysteresis filtering. A two-frame hysteresis requirement (approximately 66 ms at 30 fps) prevents false zone transitions caused by momentary tracking noise, without adding perceptible delay. This is simpler than the Kalman filtering sometimes

used in gaze tracking [14] and sufficient for the three-zone model.

Asymmetric zone thresholds. The calibration-derived thresholds use different margin ratios for the left (55%) and right (45%) sides, with minimum floors of 4% and 12% respectively. This accounts for the observation that most users have slightly asymmetric gaze ranges, and ensures both zones are reachable even with a skewed neutral gaze position.

LLM model choice. We use Llama 3.3 70B via the Groq API because it provides sub-second inference latency with reasonable cost, making it practical for real-time typing assistance. The system degrades gracefully without internet: local prefix matching continues to function.

C. Limitations

GazeSpeak has not yet been evaluated in a formal user study with ALS patients. The system requires a three-point calibration on each launch, which may be difficult for patients with very restricted eye movement range. The binary conversation engine depends on the LLM generating appropriate and non-harmful options, which is not guaranteed. The emergency blink alert uses a fixed threshold (five blinks in four seconds) that may need per-patient tuning to avoid false positives in patients with abnormal blink patterns. The step-based navigation model trades typing speed for robustness to tracking noise, and the system’s throughput has not been formally measured against commercial eye trackers.

VI. CONCLUSION

GazeSpeak combines four things that have mostly been studied separately in prior work — webcam-only iris track-

ing, LLM-accelerated prediction, a binary-choice conversation mode, and blink-triggered emergency alerting — into a single deployable tool that a patient can run on ordinary consumer hardware. None of its individual components is entirely new: LLM-accelerated typing, calibration-free interfaces, and BCI fallback paradigms have each been demonstrated before [1]–[3], [11]. What has been missing, and what this paper contributes, is the integration: a system a caregiver could actually install and use today, without infrared hardware or a research lab.

That gap between “demonstrated in the literature” and “usable by a patient” is also where the open questions sit. The three-point calibration, the fixed blink-alert threshold, and the step-based navigation model were all built as reasonable defaults rather than values tuned against real patients, and the paper is explicit about that in the limitations above. The next step is a controlled user study with ALS patients to measure typing speed, error rate, and user satisfaction against those defaults, followed by work on a BCI fallback channel for patients who progress to a locked-in state.

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